Adit Rahman

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Education

University of Waterloo — Waterloo, Ontario
Bachelor of Computer Science, Co-op | GPA: 3.86/4.00 — Sept. 2024 – Apr. 2028

- **Relevant Coursework**: Algorithms & Data Structures, Compilers, Databases, Functional Programming

Professional Experience

Shopify — Jan. 2026 – Apr. 2026
Software Engineer Intern — Ottawa, Ontario

- Designed and shipped a production job system using Node.js, BullMQ, Redis, and Kubernetes worker pods to process **122K+ daily tasks** with custom Prometheus metrics and OpenTelemetry instrumentation
- Migrated production tables from Vitess to YugabyteDB for Shopify's ads domain, processing **11.6M records** with zero-downtime across a system serving **18M+ ads/month**
- Unblocked a cross-team Airflow DAG involved in ad delivery within hours by diagnosing a **130K-row** data inconsistency, tracing the root cause to composite primary key duplication
- Built the Product Network app's first staging environment with Docker, Kubernetes, and Buildkite CI/CD, enabling pre-production validation and catching breaking changes before release

Shopify — Sept. 2025 – Dec. 2025
Software Engineer Intern — Ottawa, Ontario

- Refactored merchant eligibility with parallel validation and a decoupled cache layer for safe migration off an external dependency, sustaining **p50 32ms** latency across **10.2M** lookups at a **97.4%** cache hit rate
- Decoupled the ads-publisher service from the merchant model by introducing a request abstraction and updating GraphQL resolvers, unlocking onboarding for **230K** merchants previously blocked by the legacy integration
- Built centralized shop disablement logic handling 581 merchant lifecycle transitions, consolidating previously fragmented merchant lifecycle management
- Optimized the merchant earnings page with client-side data fetching, reducing initial load time by **8x**

Projects

Lock-In | *Python, PyTorch, OpenCV, Streamlit, SQLite*

- Built a real-time focus detection system using a fine-tuned ResNet trained on the State Farm distracted-driver dataset (**22K+ images**) to classify focused vs. distracted behavior from webcam input, achieving **0.79 F1**
- Implemented an **on-device inference pipeline** with PyTorch and OpenCV using temporal prediction smoothing for stable real-time feedback, with Streamlit and SQLite supporting session tracking and persistence
- Optimized the inference loop for low-latency webcam processing by reducing frame buffering overhead and batching prediction updates to maintain responsive real-time interaction

Event Aggregator | *Go, React, TypeScript, SQLite, WebSocket, Docker*

- Built a concurrent **event ingestion data pipeline** in Go using goroutines for data collection, with REST APIs for hot-reloading listener configuration without restarts, sustaining **3.2K req/s** at **p95 45ms** for 50 concurrent clients
- Designed a WebSocket streaming layer delivering live events to concurrent clients with minimal latency

Chess Engine | *C++20, SDL2, WebAssembly*

- Developed a cross-platform chess engine in C++20 with SDL2 graphics rendering and a WebAssembly browser build, using object-oriented programming and Observer pattern for modular multi-display architecture
- Implemented 5 AI difficulty levels of increasing difficulty with 4-ply alpha-beta pruning, piece-square evaluation, and quiescence search, cutting evaluated states by **95%** and move computation from 4s to **~200ms**

Technical Skills

Languages: Python, C++, C, TypeScript, Go, Java, Ruby, SQL
Technologies: React, Node.js, GraphQL, PyTorch, MySQL, PostgreSQL, Redis, Docker, Buildkite, Kubernetes, Airflow, Prometheus, Grafana, OpenTelemetry, OpenCV